

Garment Pattern Maker V6 — Full CAD Suite PRD

Sep 30, 2026 · @careerproai

V6 turns today's single-shirt demo into a full apparel CAD suite with feature parity to Tukatech's published product line, built on the existing Python 3.12/FastAPI backend and TypeScript/React frontend.

1. Scope and positioning

In this PRD, "follow Tukatech 100%" means matching every software capability that tukatech.com publicly documents. It does not mean copying Tukatech's code, file formats, name, branding or assets. Tukatech's internals are not public, so each requirement is written from the published feature description and designed for this codebase.

In scope (software parity):

- TUKAcad: pattern making (TUKAdesign), grading, marker making (TUKAmark), editable measurement chart, digitizing, import/export, cut-data optimization, yield reports, Made to Measure, template library, built-in help, multi-language
- TUKA APM: automatic pattern making from a spec sheet
- SMARTmark: automatic nesting, matching, marker queue, yield costing
- TUKAcutPlan: cut order and material requirement planning, dye lots
- TUKA3D: pattern-based 3D fit, avatars, fabric physics, fit maps, rendering
- TUKAstudio: color separation, repeats, colorways
- TUKAStyleHub: style folders, approvals, versioning, comments, dashboards, roles

In scope (hardware interfaces only): plotters, spreaders and cutters (TUKAjet, TUKAspread, TUKAcut and its laser and rotary variants). V6 produces the files and job queues these machines consume, such as HPGL plots and cut files. It does not build machines or firmware.

Out of scope: Windows-only desktop packaging (V6 stays a web app), billing systems, and TUKAmover and floor-balancing (no public product page was reachable).

2. Current baseline (V5)

V5 is a working demo that drafts one garment (men's regular long-sleeve shirt, 8 pieces) from an XLSX workbook and a tech-pack PDF. It then grades S-3XL, nests a marker and exports SVG, PDF and JSON. The code is about 5,900 lines of Python (including tests) and 2,200 lines of TypeScript (including tests).

Layer	Key files	What it does today	Main limit vs parity
Domain	<code>domain/drafting.py</code> , <code>geometry.py</code> , <code>catalog.py</code> , <code>profiles.py</code>	Fixed formulas build 8 pieces; quadratic curves at 24 steps; one profile <code>demo_v1</code>	No pattern editor, one garment, no grade rules
Parsing	<code>infrastructure/parsers.py</code> , <code>pdf_parser.py</code> , <code>formulas.py</code>	XLSX with safe formula evaluation and cell provenance; searchable-PDF text extraction	Shirt-specific regexes; no OCR; no DXF import
Requirements	<code>application/requirements.py</code> , <code>calibration.py</code>	Blocks generation on units, profile, review, conflicts, missing values	Hard-coded to shirt keys
Workflow	<code>application/state.py</code> , <code>pattern_workflow.py</code> , <code>reviews.py</code>	11-state machine, 6 review gates with fingerprints	Single user, no roles
Validation	<code>infrastructure/validation.py</code> , <code>seam_checks.py</code>	Structure, outline, seam-length, source-dimension checks	Shirt-specific checks

Layer	Key files	What it does today	Main limit vs parity
Marker	<code>marker.py</code> , <code>marker_pack.py</code> , <code>marker_search.py</code>	Row first-fit on bounding boxes, ≤20 garments, 0° grain only, utilization %	No true-shape nesting, rotation, flip, matching
Export	<code>exports.py</code> , <code>marker_exports.py</code>	SVG, PDF preview reports, JSON round-trip	No DXF-AAMA/ASTM, HPGL or cut files
Persistence	<code>repository.py</code> , <code>migrations.py</code> , <code>projections.py</code>	SQLite JSON snapshot plus normalized tables (schema v3), audit per save	One JSON blob per project; no multi-user
AI	<code>ai_provider.py</code> , <code>assistant_policy.py</code>	Offline regex intents; proposals need confirmation	No cloud provider
Frontend	<code>App.tsx</code> , <code>useAppController.ts</code> , <code>PatternCanvas.tsx</code> , <code>MarkerPanel.tsx</code>	Guided workflow, SVG viewer with pan and zoom, piece selection	Viewer, not an editor

V5 strengths carry forward unchanged: deterministic geometry, provenance on every value, human confirmation before AI actions, and review gates. V6 builds on them rather than replacing them.

3. Parity matrix

Of 41 published capabilities, V5 fully covers 1, partly covers 14 and lacks 26. Phases refer to the roadmap in section 13.

Capability (as published)	Tukatech product	V5 status	V6 phase
Digital pattern drafting: points, lines, curves, darts, pleats, notches, drill holes	TUKAcad / TUKAdesign	Missing	P1
Pattern modification: walk, true seams, slash-and-spread, mirror, fold	TUKAcad / TUKAdesign	Missing	P1
Seam allowance per edge, corner types	TUKAcad / TUKAdesign	Partial	P1
Custom grade rules to standardize brand fit	TUKAcad	Missing	P1
Regenerate each size from spec values	TUKA APM	Done	Keep
Editable measurement chart	TUKAcad CPE	Partial	P1
Unlimited pieces and sizes per file	TUKAcad CPE	Partial	P1
Half-size grading	TUKAcad	Missing	P1
Angle grading for curve control	TUKAcad	Missing	P1
Data import/export (DXF-AAMA/ASTM)	TUKAcad CPE	Missing	P1
Digitizing paper patterns	TUKAcad CPE	Missing	P1
All languages in one install (incl. Bangla)	TUKAcad	Missing	P1
Built-in video help	TUKAcad	Missing	P2
Custom size range	TUKA APM	Partial	P2
Silhouette library (t-shirts, trousers, polos, blocks; men and women)	TUKA APM	Partial	P2
Visual guide for each point of measure	TUKA APM	Missing	P2
One-click graded pattern ready for costing marker	TUKA APM	Partial	P2
Made to Measure: import customer measurements, nearest size, fine-tune	TUKAcad	Missing	P2

Capability (as published)	Tukatech product	V5 status	V6 phase
Fitted 2D block template library	TUKAcad	Partial	P2
Interactive marker making	TUKAmark	Missing	P3
Automatic nesting with no time limit	SMARTmark	Missing	P3
Grain, rotation, flip, buffers	SMARTmark	Partial	P3
Plaid and stripe matching, grouping, blocking	SMARTmark / TUKAcad	Missing	P3
Yield reports with costing	SMARTmark	Partial	P3
Half run, nesting blocks	SMARTmark	Missing	P3
PDF plot format	SMARTmark / TUKAcutPlan	Partial	P3
Cut data optimization, cut path	TUKAcad CPE / TUKAcut	Missing	P3
Cut order and material requirement planning	TUKAcutPlan	Missing	P3
Dye lots, fabric rolls, shading control	TUKAcutPlan	Missing	P3
Plotter output (HPGL) and job queue	TUKAjet	Missing	P3
Style folder: brand, season, category, gender, fit	TUKAStyleHub	Partial	P4
Stage approvals recorded per version	TUKAStyleHub	Partial	P4
Comments per version, notifications	TUKAStyleHub	Missing	P4
Change log, every version downloadable	TUKAStyleHub	Partial	P4
Roles, vendor accounts, admin, centralized marker queue (Net-Q)	TUKAStyleHub / SMARTmark	Missing	P4
Six rigged avatars from body scans	TUKA3D	Missing	P5

Capability (as published)	Tukatech product	V5 status	V6 phase
Fabric physics library and swatch testing	TUKA3D	Missing	P5
Fit maps: stretch, warp, weft, pressure, x-ray, tension	TUKA3D	Missing	P5
Trims, Pantone colors, prints, lights, cameras, rendering	TUKA3D	Missing	P5
Color separation and repeats	TUKAstudio	Missing	P5
Colorways saved in the master file	TUKAstudio	Partial	P5

4. Technology constraints

All new code stays in the two existing languages: Python 3.12 for the backend and compute engines, and TypeScript for the frontend. No C#, C++, Java or Rust services are added. Native speed comes only from Python packages with compiled wheels.

Area	Keep	Add (same language)	Why
API	FastAPI, Pydantic v2, Uvicorn	Typed response models replacing <code>ObjectResponse / ListResponse</code> , WebSocket progress channel	Typed contracts; long nesting jobs need progress
Persistence	SQLAlchemy 2.x	PostgreSQL for multi-user (SQLite remains for single-user), Alembic migrations, object storage for files	StyleHub needs concurrent users and file history
Geometry	Shapely 2.x, project-owned primitives in <code>domain/geometry.py</code>	NumPy, ezdxf (DXF-AAMA/ASTM read/write), pyclipper (offsets)	Pattern editing, seam corners, DXF interchange

Area	Keep	Add (same language)	Why
Nesting	<code>marker_search.py</code> strategy interface	No-fit-polygon nesting in Python with NumPy; optional OR-Tools for cut planning	True-shape nesting and cut order optimization
Jobs	none	Background worker (RQ or Celery with Redis)	"No time limit" nesting and batch grading off the request thread
Parsing	openpyxl, pdfplumber	OCR adapter behind the existing <code>AttributeExtractor</code> port	Scanned tech packs
Imaging	none	Pillow and scikit-image for color separation and repeats	TUKAstudio parity
3D	none	Frontend: three.js (TypeScript); backend: NumPy cloth solver or a WebAssembly solver called from TypeScript	TUKA3D parity in the browser
Frontend	React 19, Vite, Vitest, Playwright, lucide-react	React Router, TanStack Query, react-i18next, a canvas/SVG CAD editor module	Multi-page suite, caching, languages, editing
Auth	none (local)	OAuth2/OIDC via FastAPI security, role-based access	Roles and vendor accounts

The hexagonal layering in `markdown/docs/06_SYSTEM_ARCHITECTURE.md` stays: domain code never imports FastAPI, SQLAlchemy, React or AI SDKs.

5. Requirements: pattern CAD, APM, grading, Made to Measure

The core change is a garment-agnostic pattern model. Pieces become editable objects made of points, curves and annotations rather than fixed shirt outputs. `drafting.py` becomes one profile among many.

5.1 Pattern model (backend `domain/pattern/`)

ID	Requirement	Acceptance
PM-01	A piece holds ordered boundary segments (line, cubic Bézier, arc), internal lines, notches, drill holes, grainline, fold line, labels and cut quantity (self, mirrored pair, on fold)	Round-trip through JSON and DXF keeps every element within 0.01 cm
PM-02	Each point has a stable id and optional grade-rule reference	Editing a point keeps its id; grading resolves by id
PM-03	Seam allowance is per edge with corner styles: mitre, square, reverse, fold-back	Cut outline is valid (<code>Shapely is_valid</code>) for all four corner types
PM-04	A style holds unlimited pieces and sizes	200 pieces × 30 sizes loads and renders in under 3 s
PM-05	Every edit is a Command with undo and redo (extends <code>application/commands.py</code>)	100-step undo restores identical geometry hashes

5.2 Pattern CAD editor (frontend `src/cad/`)

ID	Requirement	Acceptance
CAD-01	Draft tools: point, line, curve, rectangle, circle, offset, parallel, perpendicular, intersection	Each tool has a keyboard shortcut and snaps to points, midpoints and edges
CAD-02	Modify tools: move, rotate, mirror, split, join, trim, extend, smooth curve	Invalid results are blocked with a message, never saved
CAD-03	Garment tools: dart create/rotate/close, pleat, tuck, slash-and-spread, add fullness, fold/unfold	Dart rotation preserves seam length within 0.01 cm
CAD-04	Walk and true seams: walk one edge along another, show length difference, add notches while walking	Sleeve cap vs armhole ease shown live in cm
CAD-05	Measure: distance, curve length, angle, piece area and perimeter	Values match backend within 0.01 cm

ID	Requirement	Acceptance
CAD-06	Rulers, grid, layers, zoom to 1:1 on screen after calibration	10 cm on screen measures 10 cm $\pm$ 1 mm
CAD-07	Geometry math runs in the backend; the editor sends Commands and renders results	No geometry decision is made only in TypeScript

5.3 Grading (domain/grading/)

ID	Requirement	Acceptance
GR-01	Grade rule tables: per point, x/y delta per size step, stored in a reusable rule library per brand	A rule table graded on one style applies to another sharing point names
GR-02	Keep V5 mode <code>REGENERATE_PER_SIZE</code> beside new <code>DELTA_GRADE</code>	Both modes available per style
GR-03	Half sizes and custom size ranges (any labels, any count)	4, 4½, 5 ... grades without errors
GR-04	Angle grading: deltas along a local axis defined by two points	Lingerie cup curve keeps shape across sizes
GR-05	Nest view of all sizes, stacked on a chosen point	Matches the V5 overlay but for any style
GR-06	Grade validation reports non-monotonic growth and seam mismatches per size	Extends <code>validate_grades</code>

5.4 Measurement chart and APM (application/apm/)

ID	Requirement	Acceptance
APM-01	Editable measurement chart: points of measure (POM) $\times$ sizes, tolerances, units cm/inch	Replaces the hard-coded 35-code <code>MAPPING</code> with per-template POM lists

ID	Requirement	Acceptance
APM-02	Silhouette library: men's and women's t-shirt, polo, woven shirt, trousers, basic blocks (bodice, skirt, sleeve, pant)	Each template generates valid pieces for every size in the sample chart
APM-03	Each POM shows a visual guide on a garment sketch	Clicking a POM highlights its location
APM-04	Four-step flow: size range → template → POM values → generate graded pattern	Output opens directly in the marker module
APM-05	Import POMs from XLSX and tech-pack PDF using the V5 parsers and requirement gates	The V5 demo shirt still passes its golden tests

5.5 Made to Measure and digitizing

ID	Requirement	Acceptance
MTM-01	Import customer measurements (CSV/XLSX), pick the nearest size by weighted POM distance	Chosen size and per-POM differences are shown
MTM-02	Apply alteration rules to the nearest size to fine-tune	Altered pattern passes validation and records the rules used
DIG-01	Digitize from a photo of a paper pattern on a calibration board, plus DXF import from digitizer software	Scale error under 0.3% on the reference board
DIG-02	Point-by-point manual trace mode as fallback	Traced piece is editable like a drafted one

6. Requirements: markers, nesting, cut planning, machine output

V5's row packer (`marker_pack.py`) stays as the fast "quick estimate" strategy. New strategies plug into the same `search_marker` interface in `marker_search.py` .

6.1 Interactive marker making (TUKAmark equivalent)

ID	Requirement	Acceptance
MK-01	Marker setup: fabric width, length limit, selvage margins, face-up/face-down, one-way or two-way, tubular	Stored per marker and shown in reports
MK-02	Drag pieces from a piece bin onto the fabric; overlap and bounds shown live	Placement is refused if it overlaps beyond tolerance
MK-03	Per-piece rules: allowed rotations (0/90/180/any), flip, tilt tolerance in degrees, buffer (uniform or per edge)	Rules enforced for manual and automatic placement
MK-04	Unlimited marker dimensions and garments (remove V5's 20-garment and 500 cm caps for Professional edition)	200-garment marker opens in under 5 s

6.2 Automatic nesting (SMARTmark equivalent)

ID	Requirement	Acceptance
NS-01	True-shape nesting with no-fit polygons, not bounding boxes	Beats V5 row packer utilization by ≥ 8 points on the demo shirt at 150 cm
NS-02	Runs as a background job with no time limit; user can stop at any time and keep the best result; progress streams to the UI	Best-so-far marker is valid at every checkpoint
NS-03	Half-run option and nesting blocks (regions that restrict or group pieces)	Pieces never leave their block
NS-04	Plaid and stripe matching: repeat size x/y, match points per piece, pieces snap to repeat lines	Matched pieces align to within 1 mm of the repeat
NS-05	Grouping: keep pieces of one garment or size together	Group bounds contain all members
NS-06	Deterministic seed for tests (keeps V5 principle)	Same seed and inputs give the same marker

6.3 Cut planning (TUKAcutPlan equivalent)

ID	Requirement	Acceptance
CP-01	Enter an order: sizes $\times$ colors $\times$ quantities, table length, max plies, max marker length	Stored as a cut order
CP-02	Generate lays: which sizes go in each marker and how many plies, minimizing fabric and number of markers	Planned quantity equals ordered quantity; shortfall or over-cut reported
CP-03	Material requirement: total fabric per color and per roll, with end loss and splicing allowance	Totals reconcile with marker lengths $\times$ plies
CP-04	Dye lots and rolls: assign orders to lots, track roll usage, mark fabric layers	No lay mixes dye lots unless allowed

ID	Requirement	Acceptance
CP-05	Marker queue listing orders with length, width and efficiency	Sortable and exportable

6.4 Reports and machine output

ID	Requirement	Acceptance
OUT-01	Yield report: fabric per garment, utilization, waste, cost per garment from fabric price	Values match marker geometry to 0.1%
OUT-02	Plot output: HPGL/2 for pattern and marker plotting, PDF plot at 1:1 tiled on A4/Letter and full-width	10 cm test square plots at 10 cm $\pm$ 0.5 mm
OUT-03	Pattern interchange: DXF-AAMA and DXF-ASTM import and export with notches, grain, drill holes, grade rules	Round-trip of 10 reference files keeps geometry to 0.01 cm
OUT-04	Cut file export for conveyor cutters (generic cut-path format with drill and notch commands) and cut-path optimization to shorten travel	Travel distance reduced vs input order; no piece cut twice
OUT-05	Spreader data: ply count, lengths, and marker image for projection	Exported per lay
OUT-06	Device job queue: send, pause, resume from checkpoint, status per plotter or cutter	Queue survives a server restart

Machine drivers are out of scope. V6 produces standard files and a queue API that a site-local agent (Python) pushes to devices over USB or network.

7. Requirements: 3D, print studio, collaboration, languages, editions

7.1 3D fit (TUKA3D equivalent)

ID	Requirement	Acceptance
3D-01	Six parametric avatars, child to adult, adjustable by body measurements, with walk and arm-raise animations	Avatar girths match input within 0.5 cm
3D-02	Sew pieces from the real 2D pattern: pair seams from pattern data, place pieces around the avatar	Unsewn or mismatched seams are listed before simulation
3D-03	Fabric library with stretch (warp, weft, bias), bend, shear, weight, thickness; add fabrics from swatch test values	Library entries are versioned; a fabric change re-runs the fit
3D-04	Fit maps: stretch, warp, weft, pressure, tension, and x-ray (garment transparent over body); frame-by-frame in motion	Maps use a shared legend in % strain or kPa
3D-05	2D↔3D link: a pattern edit updates the 3D sample	Edit-to-refit under 10 s for an 8-piece shirt
3D-06	Trims (buttons with thread), Pantone-style color picking, placement prints and logos, color/transparency/shine/texture channels	Materials render the same in viewer and still renders
3D-07	Virtual photo shoot: lights, cameras, backgrounds, still and turntable video export	PNG and MP4 export
3D-08	Browser-only 3D review for approvers with nothing to install	Works in current Chrome, Edge and Safari

The simulation runs client-side in TypeScript (three.js plus a cloth solver) so reviewers need no install; the backend stores fabric data, seam pairs and results.

7.2 Print and textile studio (TUKAstudio equivalent)

ID	Requirement	Acceptance
ST-01	Color separation: reduce scanned artwork to N colors by clicking, with automatic mode	Output palette has exactly N colors
ST-02	Repeats: full drop, half drop, brick, mirror; stripe tools; motif library; edit across repeat edges	Tiled output is seamless
ST-03	Colorways: generate many color variants from a palette; all saved in one master file	Each colorway links to the style
ST-04	Send a print to 3D and to plaid/stripe matching in the marker	Repeat size flows into NS-04

7.3 Collaboration hub (TUKAStyleHub equivalent)

ID	Requirement	Acceptance
HUB-01	A style folder holds sketch, tech pack, measurement chart, patterns, 3D samples, prints, markers, trims, fabrics; filed by brand, season, category, gender, garment type, fit	Search and filter by any field
HUB-02	Stages: concept → pattern → sample → fit → costing → final approval, with an owner per stage	Generalizes V5's six review gates and fingerprints
HUB-03	Approvals and rejections recorded against a version; send-backs counted	Approval rate and rework count per style
HUB-04	Every version downloadable; change log of who changed what and when	Extends V5 <code>audit</code> and <code>reviews</code> into first-class tables
HUB-05	Comments tied to a version; notifications to the next owner	Email and in-app notifications

ID	Requirement	Acceptance
HUB-06	Dashboards: work in progress, awaiting review, completed, styles per brand, activity per day; export to Excel and CSV	Any date window
HUB-07	Roles per department and user; external brand and vendor accounts scoped to shared styles; company admin	Vendor sees only shared styles
HUB-08	Shared marker queue across users and sites (Net-Q equivalent)	Two users on different sites see the same queue

7.4 Languages and help

ID	Requirement	Acceptance
I18N-01	All languages in one build, switchable at runtime: English, Bangla, Chinese, French, Hindi, Italian, Spanish, Thai, Vietnamese	Every UI string comes from a catalog; no hard-coded text
I18N-02	Units cm and inch everywhere, fractions for inch	Unit switch never changes stored values
HELP-01	Built-in video help per tool and a searchable help panel	Each toolbar tool links to a clip

7.5 Editions and entitlements

Editions mirror the published tier structure. Prices are a business decision and are not set here.

Edition	Pattern and grading	Marker	Import/export, digitizing, cut optimization, yield	Limits
Learning	Yes	Yes	No	14 pieces and 4 sizes per file; markers up to 59 × 120 in
Design Professional	Yes, editable measurement chart	No	No	Unlimited pieces and sizes
CAD Professional	Yes, editable measurement chart	Yes	Yes	Unlimited, including marker size
APM add-on	Automatic pattern making	—	—	Added to Design or CAD Professional

Entitlements are enforced in the backend application layer, not only hidden in the UI.

8. Data model and API additions

The single JSON snapshot in `projects.data` is retired in favor of the normalized tables V5 already projects in `projections.py`, which become the source of truth. A one-time migration (schema v4) converts existing snapshots; snapshots stay readable for rollback.

Entity	Key fields	Replaces or extends
Organization, User, Role, Membership	org, department, role, vendor scope	New (auth)
Style	brand, season, category, gender, garment type, fit, stage, owner	V5 <code>projects</code>
SizeRange, MeasurementChart, POM, POMValue	labels, tolerances, units, provenance	V5 <code>measurements</code> , <code>measurement_values</code>

Entity	Key fields	Replaces or extends
Template (APM silhouette)	POM list, drafting script version, sketch	V5 <code>profiles.py</code>
Piece, Segment, Point, Annotation	geometry, grade-rule refs, seam per edge	V5 <code>pattern_pieces</code>
GradeRuleTable, GradeRule	point, size step, dx, dy, angle axis	New
PatternVersion	style, number, input hash, validation status	V5 <code>pattern_sets</code>
Fabric	width, repeat x/y, direction, physics values, price per metre	New
Marker, Placement, NestJob	rules, utilization, status, best-so-far	V5 <code>markers</code>
CutOrder, Lay, DyeLot, Roll	sizes × colors × quantities, plies	New
DeviceJob	device, file, status, checkpoint	New
Approval, Comment, Notification	version, decision, author	V5 <code>reviews</code> , <code>audit_events</code>
Avatar, Sample3D, Print, Colorway	measurements, sim results, palette	New

API conventions: keep the `/api/v1` prefix and the V5 error envelope (`code` , `message` , `details` , `correlation_id`). New routers: `/styles` , `/charts` , `/templates` , `/pieces/{id}/commands` , `/grade-rules` , `/markers` , `/nest-jobs` (with WebSocket `/nest-jobs/{id}/progress`), `/cut-orders` , `/devices/jobs` , `/imports/dxf` , `/exports/{kind}` (adds `dxf-aama` , `dxf-astm` , `hpgl` , `cut`), `/samples3d` , `/prints` , `/approvals` , `/comments` , `/dashboards` . Every list endpoint is paginated, and every response uses a typed Pydantic model.

9. Non-functional requirements

Area	Requirement
Accuracy	Stored geometry in cm with 1e-6 coordinate tolerance (V5 <code>tolerances.py</code>); lengths reported to 0.01 cm; plots at 1:1 within ± 0.5 mm per metre
Determinism	Same inputs, rules and seed give identical geometry and marker hashes
Performance	Editor command round-trip under 150 ms at p95; grading 30 sizes $\times$ 50 pieces under 2 s; 3D refit of an 8-piece shirt under 10 s
Scale	500 concurrent users per organization; 10,000 styles; files up to 50 MB
Availability	99.5% monthly for the hosted service; single-user local mode keeps working offline
Security	OIDC login, role checks in the application layer, tenant isolation on every query, audit of every write, rate limits on uploads and nesting jobs
Privacy	Customer body measurements (MTM) encrypted at rest, deletable on request
Accessibility	WCAG 2.2 AA for all non-canvas UI; keyboard access to every CAD tool (extends V5 axe tests)
Browsers	Current Chrome, Edge, Firefox, Safari; 1366 $\times$ 768 minimum for CAD
Quality gates	pytest, Vitest and Playwright stay mandatory; ruff and mypy clean; golden geometry tests per template; DXF round-trip suite
Observability	Keep V5 correlation IDs and structured logs; add metrics for job queue length and nest duration

10. Fix-first defects (Phase 0)

The repository's own audit (`artifacts/qa/gap-audit-2026-09-10.md`) reproduced five data-integrity defects. They were not re-run for this PRD. Each gets a regression test promoted from `artifacts/qa/test_gap_audit.py` before V6 work starts.

- ☐ Measurement PATCH on `/measurements/{id}` ignores its route scope and skips range checks (−3 cm saved to sleeve length)
- ☐ Archived workbook still counts as ready and generation succeeds
- ☐ JSON import accepts invalid nested grades and markers
- ☐ JSON import returns 200 for a project that does not exist
- ☐ Grading an older pattern version uses the current seam allowance
- ☐ Align assistant context (base-pattern-first) with `App.tsx` (grade-first) version selection
- ☐ Replace generic `ObjectResponse` / `ListResponse` wrappers with typed models on touched routes

11. Risks and open questions

Risk	Impact	Mitigation
Tukatech's exact tool behavior and file formats are not public	Parity claims could be wrong in detail	Test against published feature text plus pattern-maker user trials; never claim file compatibility with proprietary formats
Using Tukatech names, logos or look in the product	Trademark and passing-off exposure	Own brand and UI; Tukatech named only in internal planning
True-shape nesting in pure Python is slow	Weak marker efficiency	NumPy vectorization, background jobs, profile early; accept a compiled-wheel dependency if needed
Browser cloth simulation accuracy	3D fit not trusted	Calibrate fabric values against physical swatch tests; label 3D as visual guidance

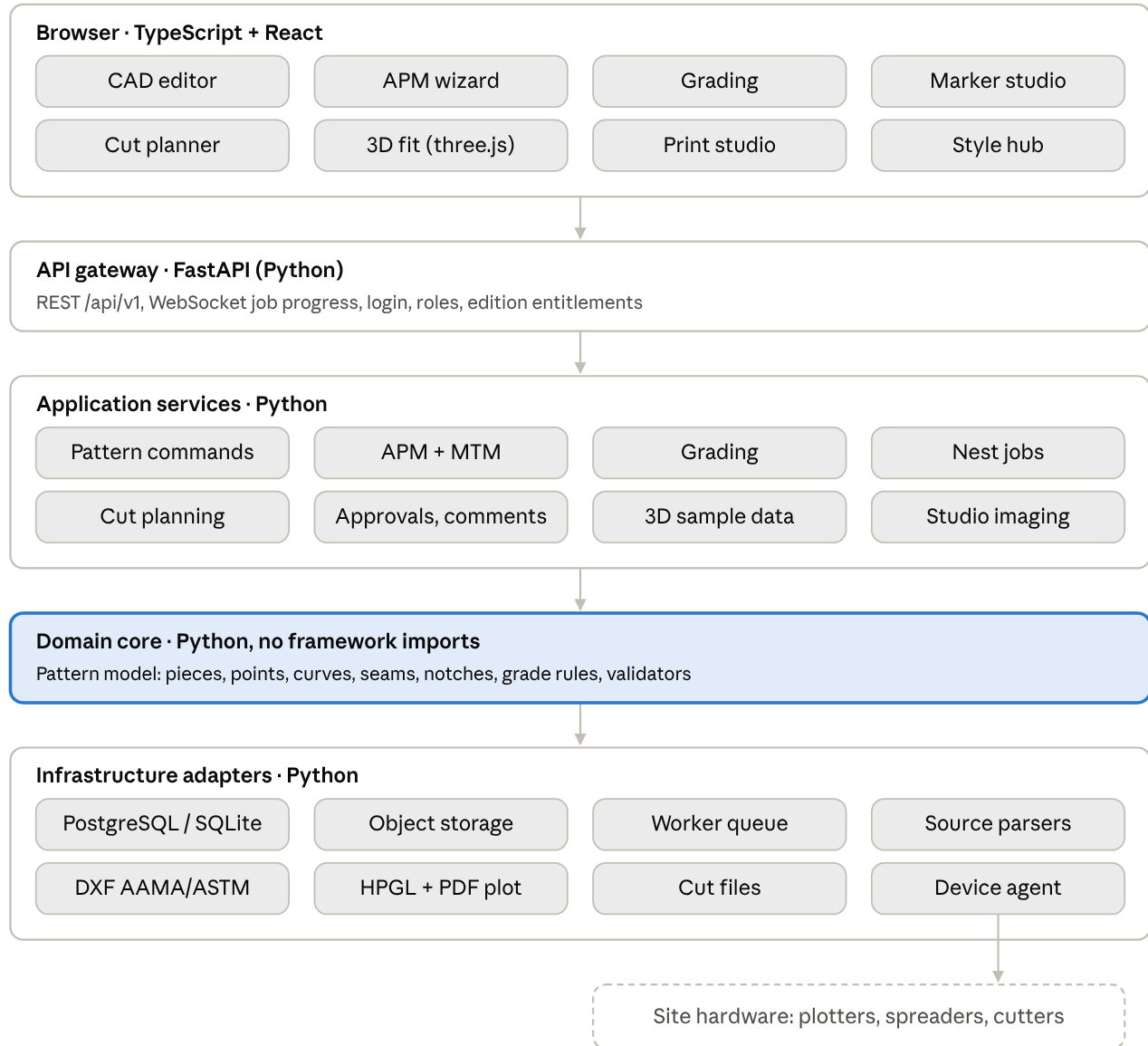
Risk	Impact	Mitigation
DXF-AAMA/ASTM dialect differences across CAD vendors	Import failures	Build a corpus of real files from users before P1 ends
Scope is about five products	Long timeline	Ship each phase as a usable release; editions let early phases earn revenue

Open questions

- ☐ Which garment templates come first after the shirt: t-shirt, polo, trousers?
- ☐ Hosted SaaS only, or also an on-premise install for factories?
- ☐ Which plotter and cutter models must the device agent support at launch?
- ☐ Are production drafting blocks and grade rules available from a client to calibrate against?

12. Target architecture

Every module runs on one Python pattern core

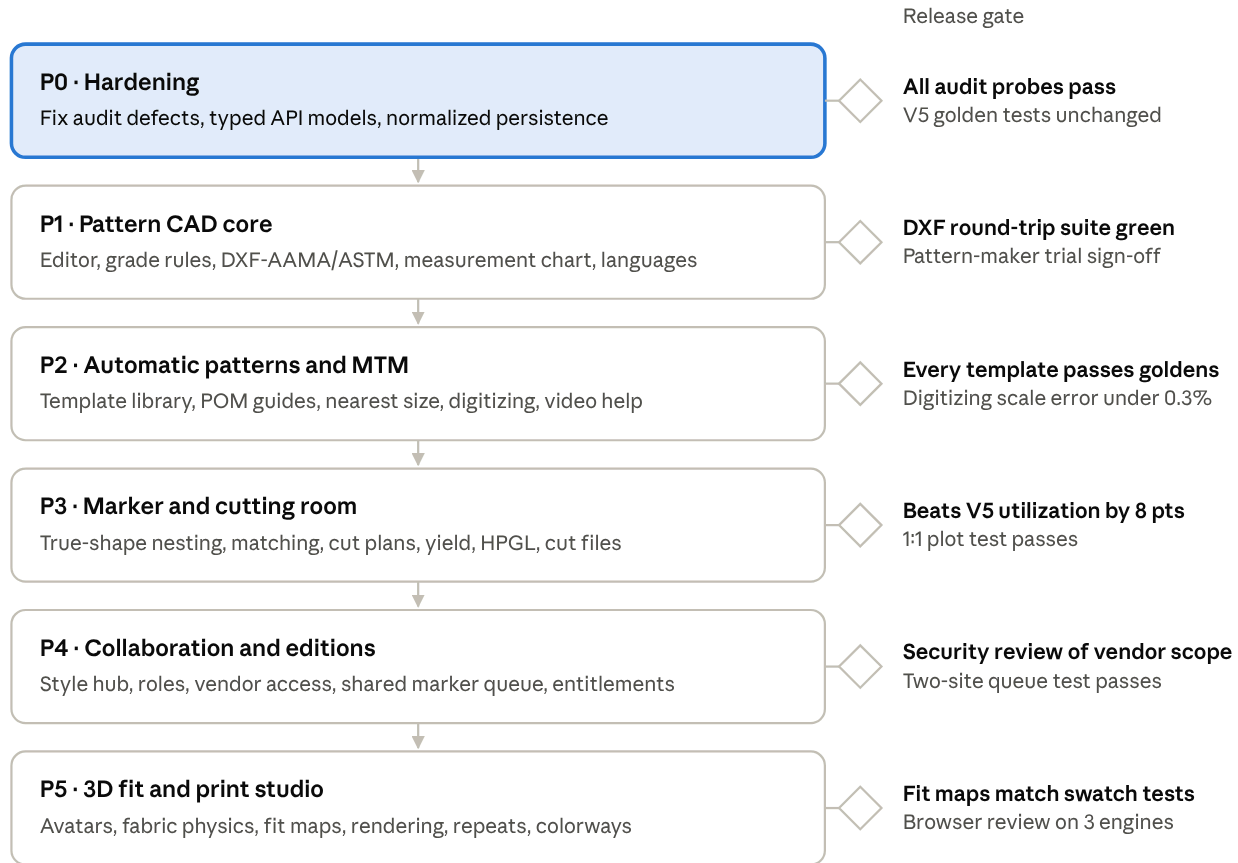


target architecture · 5 layers

React modules render and send commands; every geometry, grading and nesting decision runs in Python over one shared pattern model. Machines receive files through a site-local agent.

13. Roadmap

Six phases, each released behind a gate



roadmap · 6 phases, not to scale

Phase 0 starts first because V6 builds on V5's data integrity. Each later phase ships as a usable release and edition. Durations are not set until team size and the open questions in section 11 are answered.

14. AI features

AI reads, suggests and explains; deterministic Python code still draws every pattern. This keeps V5's rule: an AI result is a proposal that is schema-checked, shown as a preview, and applied only after the user confirms (`assistant_policy.py` , `ai_provider.py`).

ID	Feature	What the user does	Guardrail / acceptance	Phase
AI-01	Measurement chart from any tech pack: read PDFs, scans and photos (OCR + language model) into points of measure and sizes	Uploads a buyer's spec sheet	Every value shows its source location and confidence; low-confidence values block generation until confirmed	P2
AI-02	Label mapping across buyers and languages, including Bangla ("sleeve length" → sleeve length)	Reviews suggested mappings	Mapping is saved per buyer; no value changes without confirmation	P2
AI-03	Sketch or photo to template: detect garment type and details (collar, cuff, pocket, fit) and preselect the template	Uploads a flat sketch or product photo	Shows detected details with confidence; user can change any choice	P2
AI-04	Measurement checks before drafting: flag typos and outliers against the size-range trend and normal body proportions	Sees warnings in the chart	Warning only, never an automatic fix	P1
AI-05	Pattern commands in plain language: "lengthen sleeve 2 cm on all sizes", "add 1 cm seam allowance to side seams"	Types or speaks a request	Converted to editor Commands; preview of the change; undoable	P1
AI-06	Grade rule suggestions from the measurement chart	Approves or edits proposed rules	Graded sizes must match the chart within 0.01 cm	P1

ID	Feature	What the user does	Guardrail / acceptance	Phase
AI-07	Fit correction advisor: reads 3D fit maps or fit-session photos and comments, suggests pattern fixes (for example, lower armhole 1 cm)	Chooses which fixes to apply	Each fix is a previewed Command; re-simulation shows before and after	P5
AI-08	Fabric consumption and cost estimate from the tech pack before a pattern exists	Enters fabric width and price	Estimate labelled as such; replaced by marker yield once available	P3
AI-09	Nesting seed from past markers to reach good utilization faster	Nothing; runs inside nest jobs	Final marker still passes overlap, bounds and grain checks	P3
AI-10	Print studio help: color reduction, repeat detection, colorways from a text prompt or palette	Picks from generated options	Output palette and repeat size are exact and editable	P5
AI-11	Assistant that explains tools, warnings and validation results in the user's language	Asks questions in chat	Answers cite the relevant piece, rule or help clip	P2
AI-12	Style hub summaries: comment threads, approval blockers, weekly status per brand	Reads the summary	Links back to every comment it summarizes	P4

Rules for all AI features

- The provider stays behind the `AIProvider` port; the app works fully with AI switched off.
- Customer data leaves the server only when the organization opts in to a cloud provider.
- Every AI proposal logs model, prompt version, confidence and the user's decision in the audit trail.

- AI never writes pattern geometry directly; only validated Commands do.

Sources

- [TUKAcad](#)
- [TUKA APM](#) and [TUKA APM overview](#)
- [SMARTmark](#)
- [TUKAcutPlan](#)
- [TUKA3D](#)
- [TUKAStyleHub](#)
- [TUKAstudio](#)
- [TUKAjet](#), [TUKAspread](#), [TUKAcut](#)
- [Tukatech product list](#)
- Codebase: `Garment_Patterns` (backend, frontend, `markdown/docs`, `artifacts/qa`), scanned 2026-09-30